Module 8: MAT 181 Chapter 5 Practice Problems

Name_____ Date: _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Identify the given random variable as being discrete or continuous.

1) The height of a randomly selected student
   A) Continuous B) Discrete

2) The number of phone calls between New York and California on Thanksgiving day
   A) Continuous B) Discrete

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Determine whether the following is a probability distribution. If not, identify the requirement that is not satisfied.

3) In a certain town, 20% of adults have a college degree. The accompanying table describes the probability distribution for the number of adults (among 4 randomly selected adults) who have a college degree.

x	P(x)
0	0.4096
1	0.4096
2	0.1536
3	0.0256
4	0.0016

4) A police department reports that the probabilities that 0, 1, 2, 3, and 4 car thefts will be reported in a given day are 0.183, 0.311, 0.264, 0.150, and 0.064, respectively.

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Find the mean of the given probability distribution.

5) A police department reports that the probabilities that 0, 1, 2, and 3 burglaries will be reported in a given day are 0.53, 0.43, 0.03, and 0.01, respectively.
   A) $\mu = 0.25$ B) $\mu = 1.05$ C) $\mu = 1.50$ D) $\mu = 0.52$

6) In a certain town, 40% of adults have a college degree. The accompanying table describes the probability distribution for the number of adults (among 4 randomly selected adults) who have a college degree.

x	P(x)
0	0.1296
1	0.3456
2	0.3456
3	0.1536
4	0.0256

A) $\mu = 1.73$ B) $\mu = 2.00$ C) $\mu = 1.60$ D) $\mu = 1.50$

Provide an appropriate response. Round to the nearest hundredth.

7) Find the standard deviation for the given probability distribution.

x	P(x)
0	0.12
1	0.07
2	0.21
3	0.30
4	0.30

A) $\sigma = 1.35$ B) $\sigma = 2.90$ C) $\sigma = 1.30$ D) $\sigma = 1.70$

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Provide an appropriate response.

8) Let the random variable x represent the number of tails in five flips of a coin. Construct a table describing the probability distribution, then find the mean and standard deviation.

x	P(x)	x*P(x)	x^2	x^2*P(x)
0				
1				
2				
3				
4				
5				

9) In a game, you have a 1/36 probability of winning $94 and a 35/36 probability of losing $8. What is your expected value?

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

10) Suppose you buy 1 ticket for $1 out of a lottery of 1,000 tickets where the prize for the one winning ticket is to be $500. What is your expected value?

A) $0.00 B) -$1.00 C) -$0.50 D) -$0.40

Determine whether the given procedure results in a binomial distribution. If not, state the reason why.

11) Rolling a single die 57 times, keeping track of the numbers that are rolled.
   A) Not binomial: there are too many trials.
   B) Not binomial: there are more than two outcomes for each trial.
   C) Not binomial: the trials are not independent.
   D) Procedure results in a binomial distribution.

12) Choosing 10 marbles from a box of 40 marbles (20 purple, 12 red, and 8 green) one at a time with replacement, keeping track of the number of red marbles chosen.
   A) Not binomial: there are too many trials.
   B) Not binomial: the trials are not independent.
   C) Not binomial: there are more than two outcomes for each trial.
   D) Procedure results in a binomial distribution.

Assume that a procedure yields a binomial distribution with a trial repeated n times. Use the binomial probability formula to find the probability of x successes given the probability p of success on a single trial. Round to three decimal places.

13) $n = 30$, $x = 12$, $p = 0.20$
   A) 0.006 B) 0.108 C) 0.003 D) 0.014

14) $n = 14$, $x = 6$, $p = 0.5$
   A) 0.275 B) 0.183 C) 0.016 D) 0.238

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Find the indicated probability.

15) In a survey of 300 college graduates, 53% reported that they entered a profession closely related to their college major. If 9 of those survey subjects are randomly selected without replacement for a follow-up survey, what is the probability that 3 of them entered a profession closely related to their college major?

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

16) Suppose that 11% of people are left handed. If 8 people are selected at random, what is the probability that exactly 2 of them are left handed?

A) 0.0121 B) 0.337 C) 0.0416 D) 0.168

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Find the indicated probability. Round to three decimal places.

17) A test consists of 10 true/false questions. To pass the test a student must answer at least 6 questions correctly. If a student guesses on each question, what is the probability that the student will pass the test?

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

18) Find the probability of at least 2 girls in 7 births. Assume that male and female births are equally likely and that the births are independent events.

A) 0.063 B) 0.938 C) 0.164 D) 0.773

Find the mean, μ, for the binomial distribution which has the stated values of n and p. Round answer to the nearest tenth.

19) $n = 38$; $p = 0.2$

A) $\mu = 7.1$ B) $\mu = 7.6$ C) $\mu = 7.9$ D) $\mu = 8.3$

Find the standard deviation, σ, for the binomial distribution which has the stated values of n and p. Round your answer to the nearest hundredth.

20) $n = 503$; $p = 0.7$

A) $\sigma = 10.28$ B) $\sigma = 13.55$ C) $\sigma = 14.40$ D) $\sigma = 7.87$

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Use the given values of n and p to find the minimum usual value $\mu - 2\sigma$ and the maximum usual value $\mu + 2\sigma$. Round your answer to the nearest hundredth unless otherwise noted.

21) $n = 93$, $p = 0.24$

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

22) $n = 246$, $p = 0.15$

A) Minimum: 31.3; maximum: 42.5
B) Minimum: -25.83; maximum: 99.63
C) Minimum: 25.7; maximum: 48.1
D) Minimum: 48.1; maximum: 25.7

Determine if the outcome is unusual. Consider as unusual any result that differs from the mean by more than 2 standard deviations. That is, unusual values are either less than $\mu - 2\sigma$ or greater than $\mu + 2\sigma$.

23) A survey for brand recognition is done and it is determined that 68% of consumers have heard of Dull Computer Company. A survey of 800 randomly selected consumers is to be conducted. For such groups of 800, would it be unusual to get 634 consumers who recognize the Dull Computer Company name?

A) Yes
B) No

24) A survey for brand recognition is done and it is determined that 68% of consumers have heard of Dull Computer Company. A survey of 800 randomly selected consumers is to be conducted. For such groups of 800, would it be unusual to get 530 consumers who recognize the Dull Computer Company name?

A) Yes
B) No